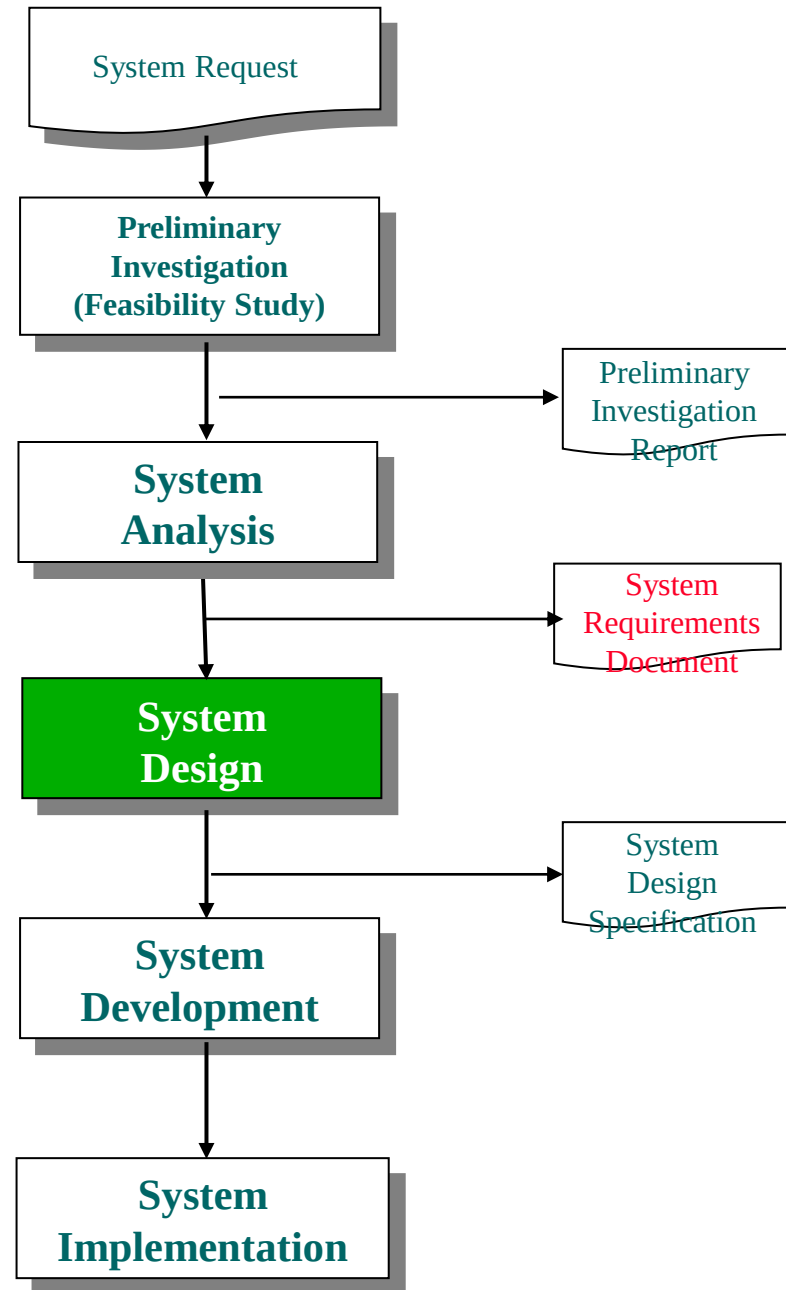


Chapter 6

System and Interface Design

SDLC



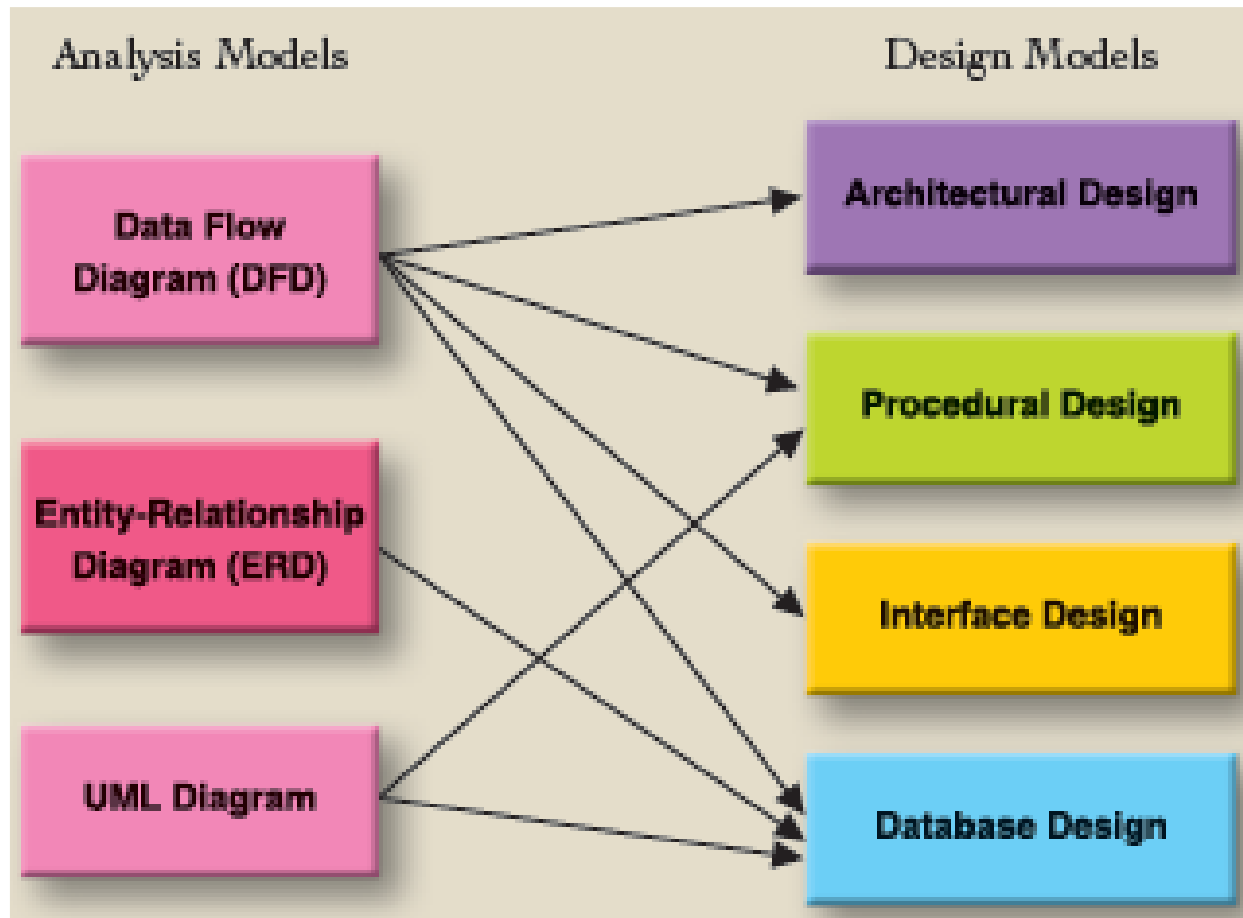
What is System Design

- Purpose : Design shows **how** the system will fulfill what the system should do.
- **Output** : Design **translates** the *System Requirements Spec* into a 'blueprint' called *System Design Specification*.
- **Person** : **Designer** details the system specifications.
- **Areas** : ISs may have some possible designs eg database, forms, security, interfaces (input, output), procedures, reports etc.

Importance of Design

- Maintenance . It provides necessary 'roadmap' and **documentation** for maintenance staff.
- Quality . Quality of design affects quality of software. A good design produces a quality software.
- Testing . Design outputs (eg Design Spec) provide the necessary **criteria** for testing of the software.
- **Communication**. It serves as a **communication** medium between the designers of sub-systems eg between Purchasing and Warehousing systems.
- **Development**. Design will be **developed** (programmed) into a software.

Overview Areas of Design



**The Analysis
Model**



**The Design
Model**

Physical and Logical Design

- Logical Design

- **Phase** : Developed during the **systems analysis** phase.
- **Focus**: Describes the **functions and features** of the system (eg. input, process, output) that must be performed.
- **Design** : Defines what functions must be carried out eg what data to input.

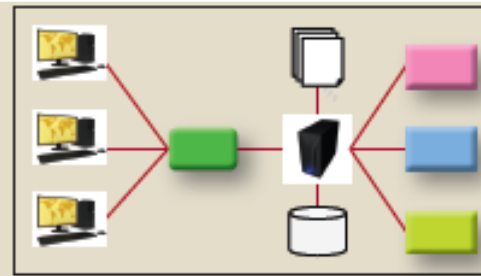
- Physical Design

- **Phase** : Developed during the **systems design** phase.
- **Focus**: Describes the **actual physical processes** of entering (eg barcode), verifying, and storing data.
- **Design** : Defines how the functions will be implemented physically eg use of barcode for input.

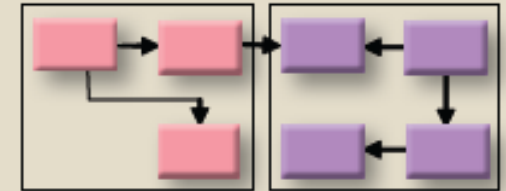
Exercise for Student

Differences	Logical Design	Physical Design
Phase	System Analysis	System Design
Focus	Describe the functions & features	Describe actual physical process
Design	Defines what must be done in the system	Describe how system will be implemented

Overview of *Models* Used in Design Stage



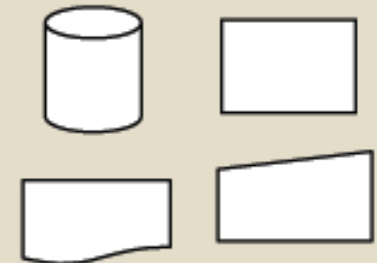
System
Architecture



System Structure



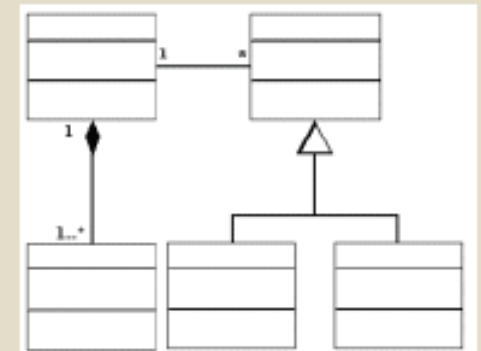
Structure Chart



System Flowchart



Report Layout



Class Diagram

System Architecture Design

1. System Architecture Design

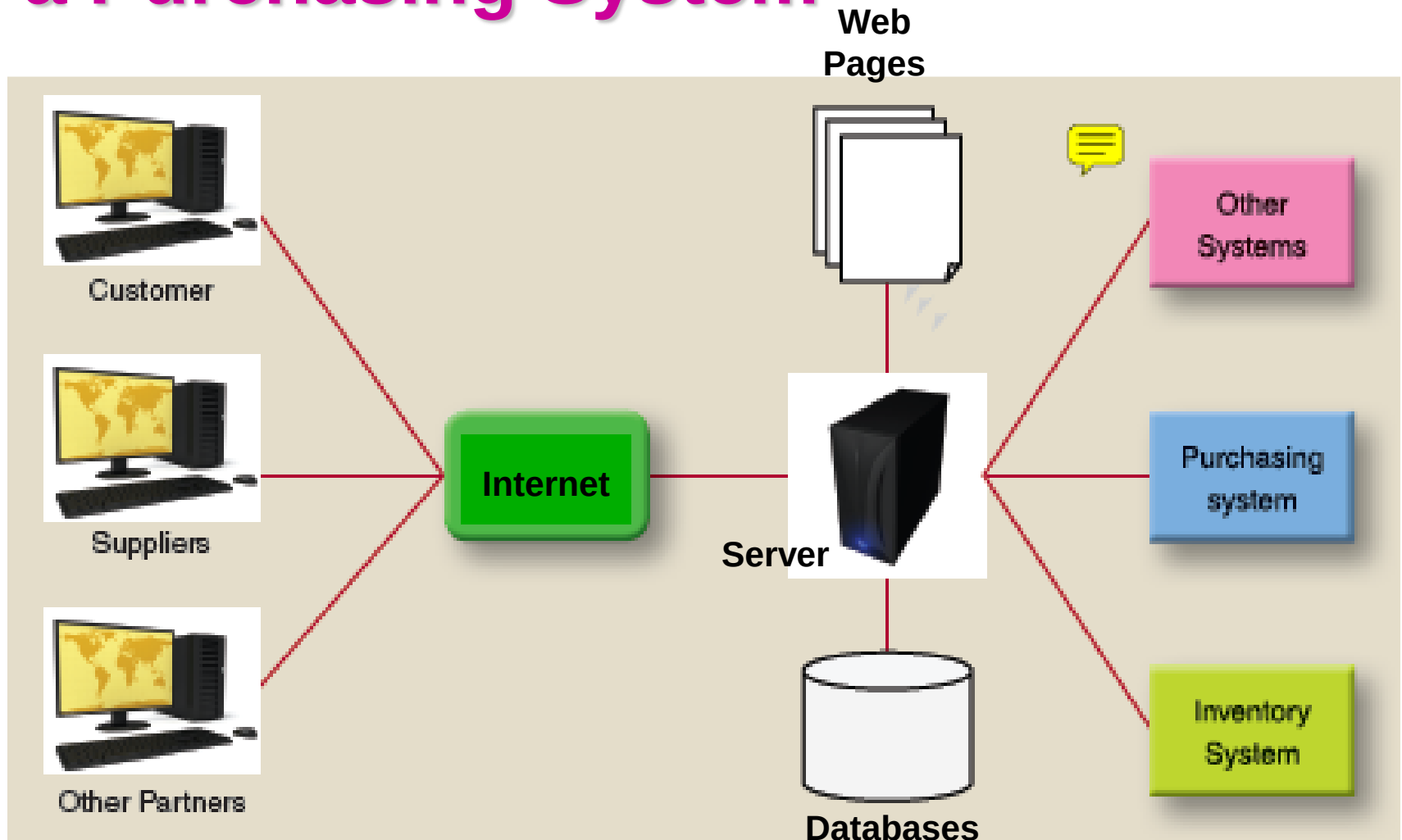
- **Meaning**

- Shows an overall **architectural** design structure of the **entire system** consisting of both internal systems, external systems and hardware (server, clients and network infrastructure), and their **relationship** between them.

- **Purpose**

- To ensure that the design of the entire system is more **structured**, and hence enhancing the development and future maintenance process.

Example 1 – Architecture Design for a Purchasing System



**Clients and
External Systems**

**Network
Infrastructure**

**Internal
Systems**

System Structure Design

2. System Structure Design

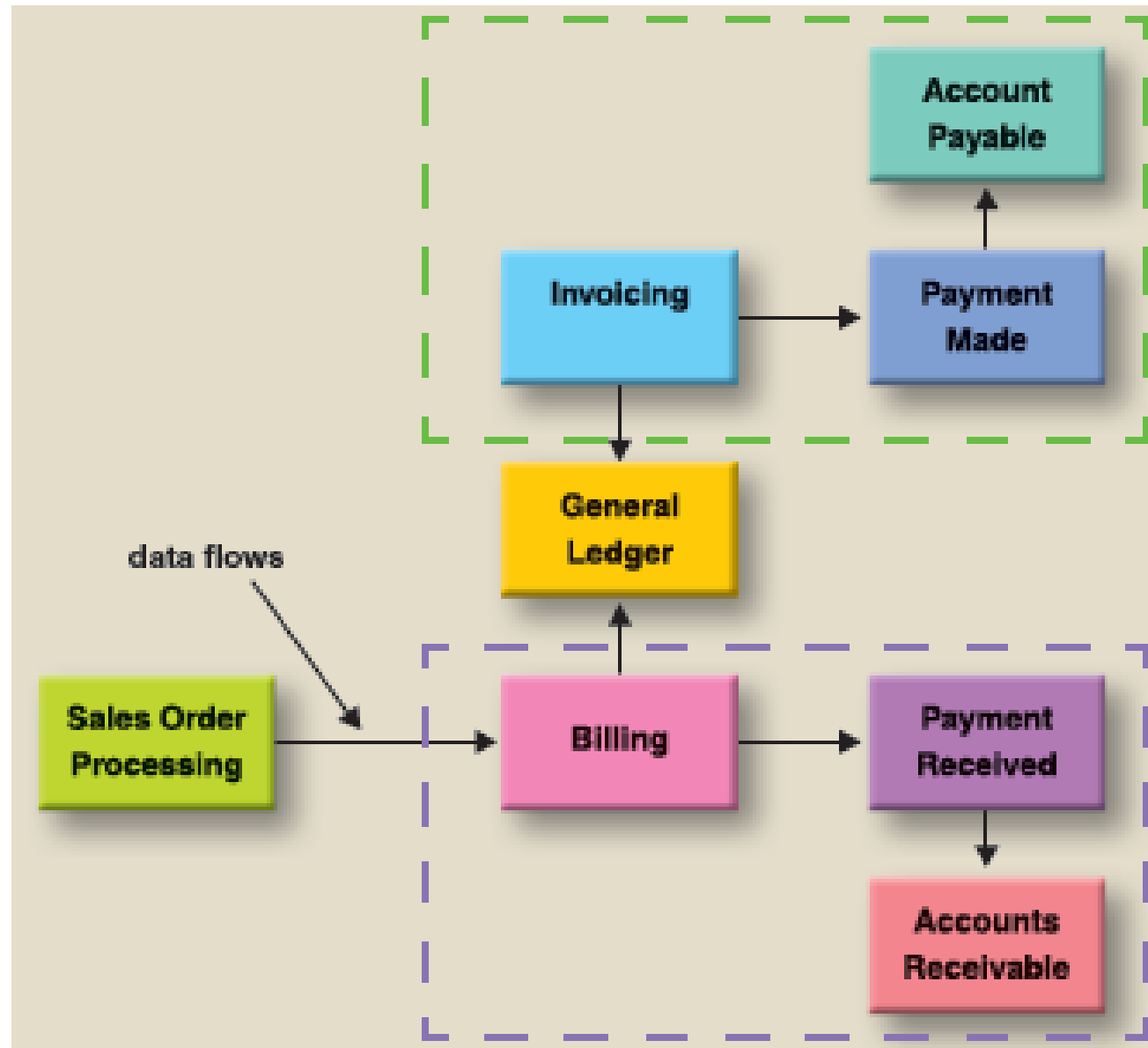
- **Meaning**

- Shows the design structure of an internal **system**, consisting of various modules and how they are related (eg. information flow) to each other.

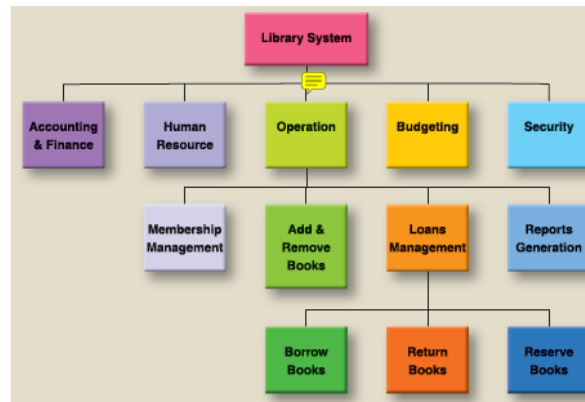
- **Purpose**

- To design a more **structured system** (eg. high modularity and low coupling) to enhance development and maintenance process.

Example 1 – Accounting System Structure



System Function Design



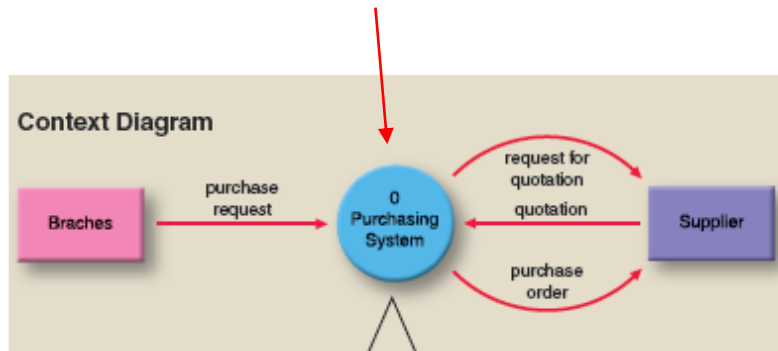
Note : The examples given below are mere **EXAMPLES**. They will **NOT** represent the actual designs of any real systems. **DO NOT** copy them into your Assignment

Functional Decomposition Diagram (Structure Charts)

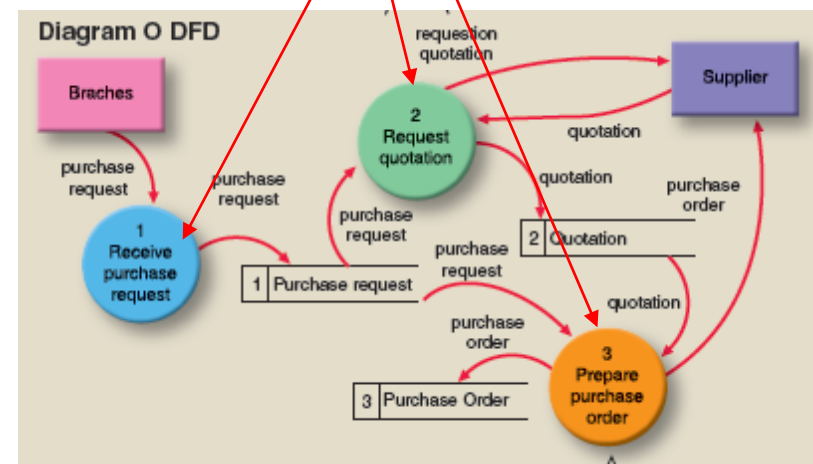


- **Design Tool** - FDD is a design tool used for a **top-down** representation of a system and its modules.
- Each **module** contains many functions or processes.
- **DFD** can be used to draw the Structure Chart. Context Diagram represents the whole **system** while the low-level DFD represents the **modules**

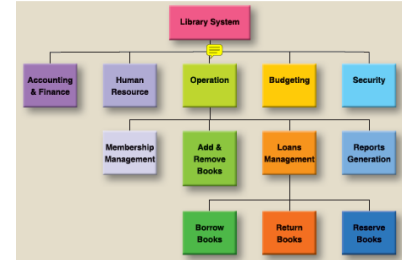
System



Modules

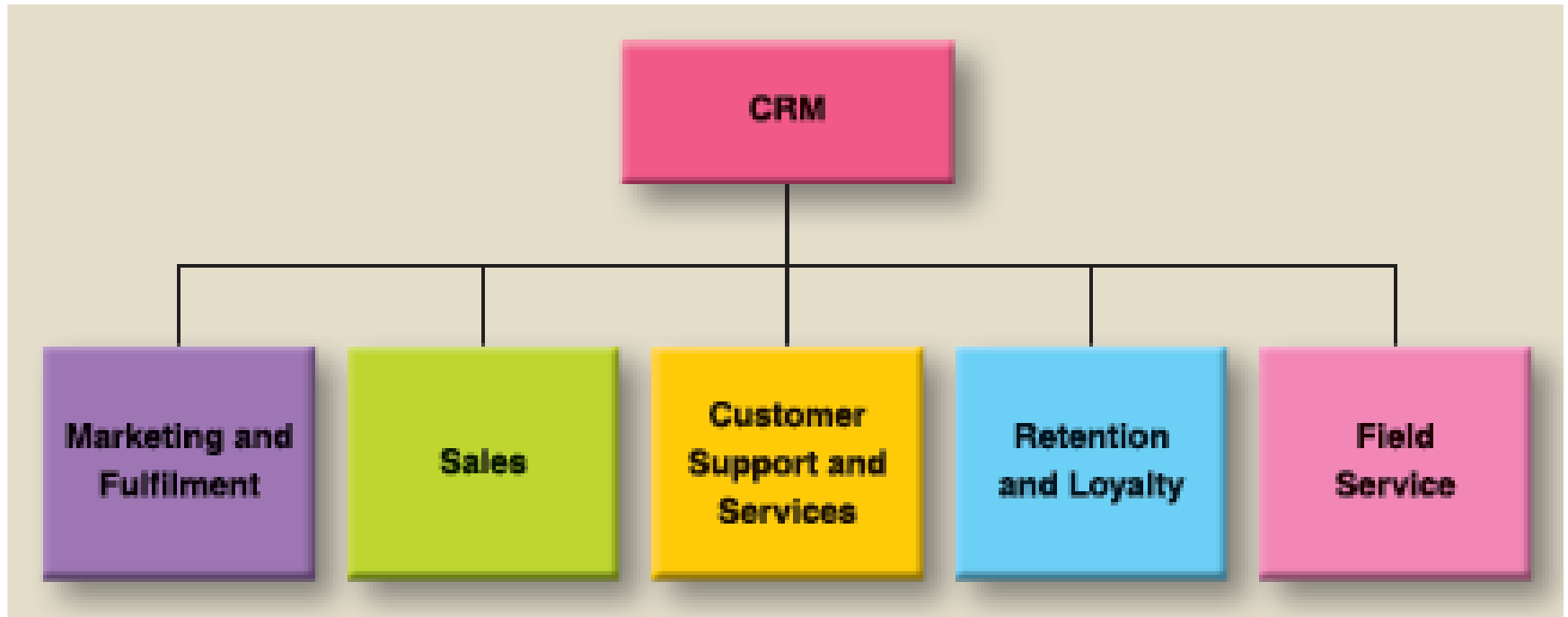


Functional Decomposition Diagram (Structure Chart)

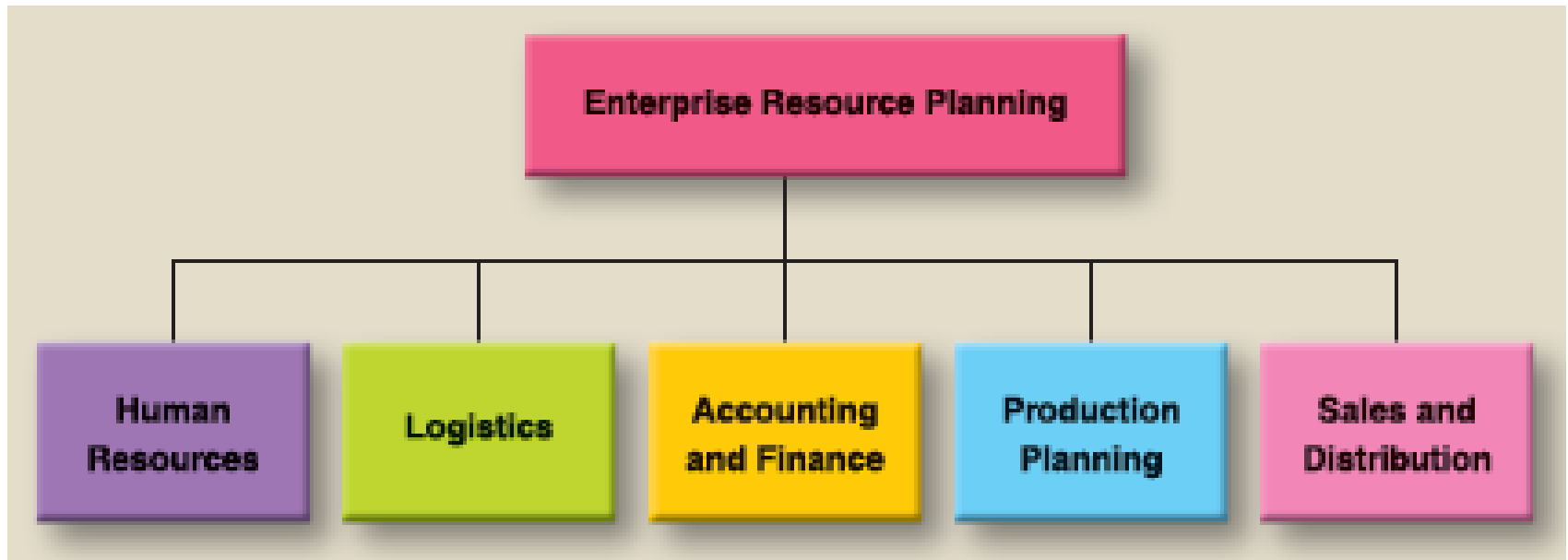


- FDDs can be used at several stages of systems development.
 - During analysis phase: FDDs are used to identify business **functions**
 - During design phase: FDDs show how the DFDs are organized into lower-level **modules** (processes).
 - During development phase: the processes can be translated into **program codes**.

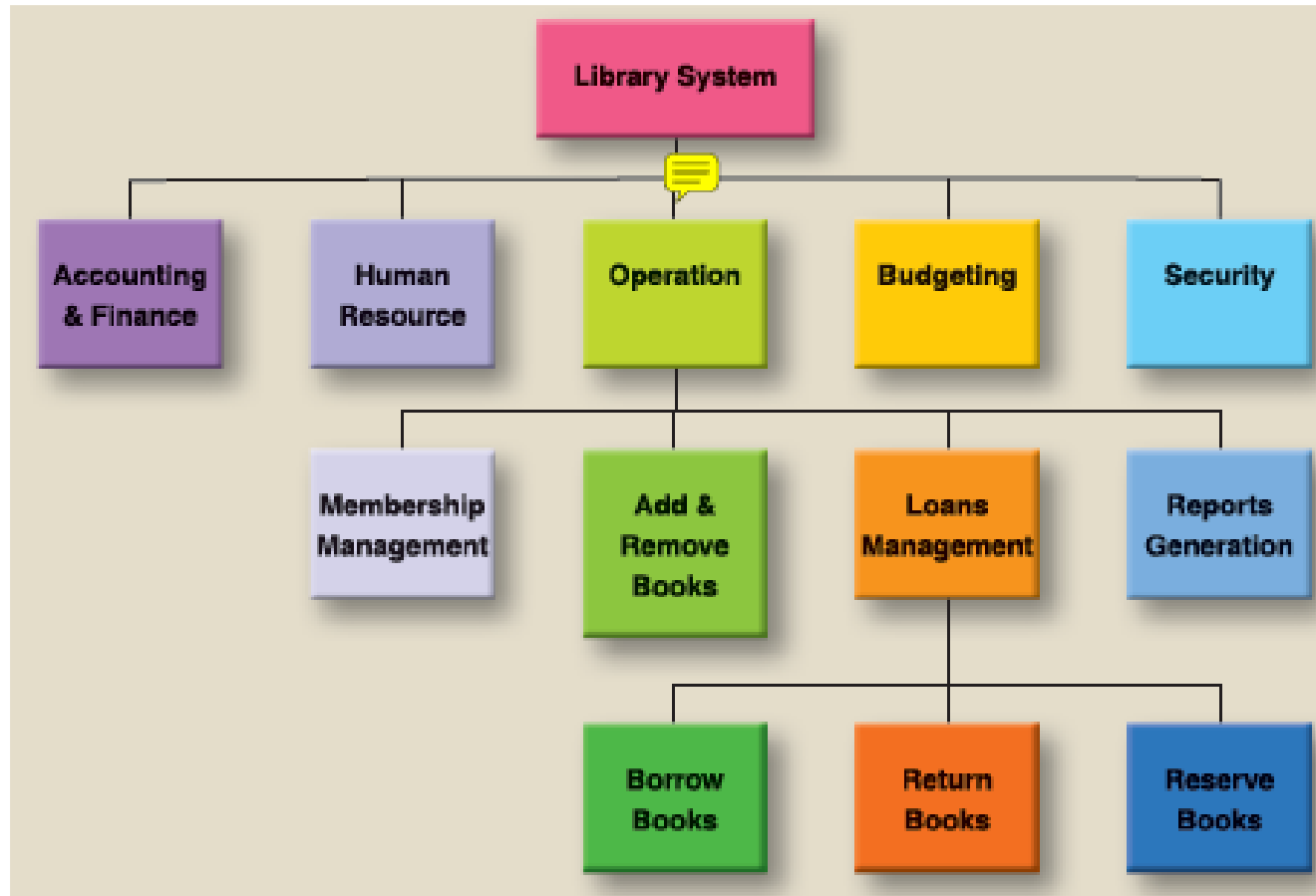
Example 1 - CRM



Example 2 - ERP

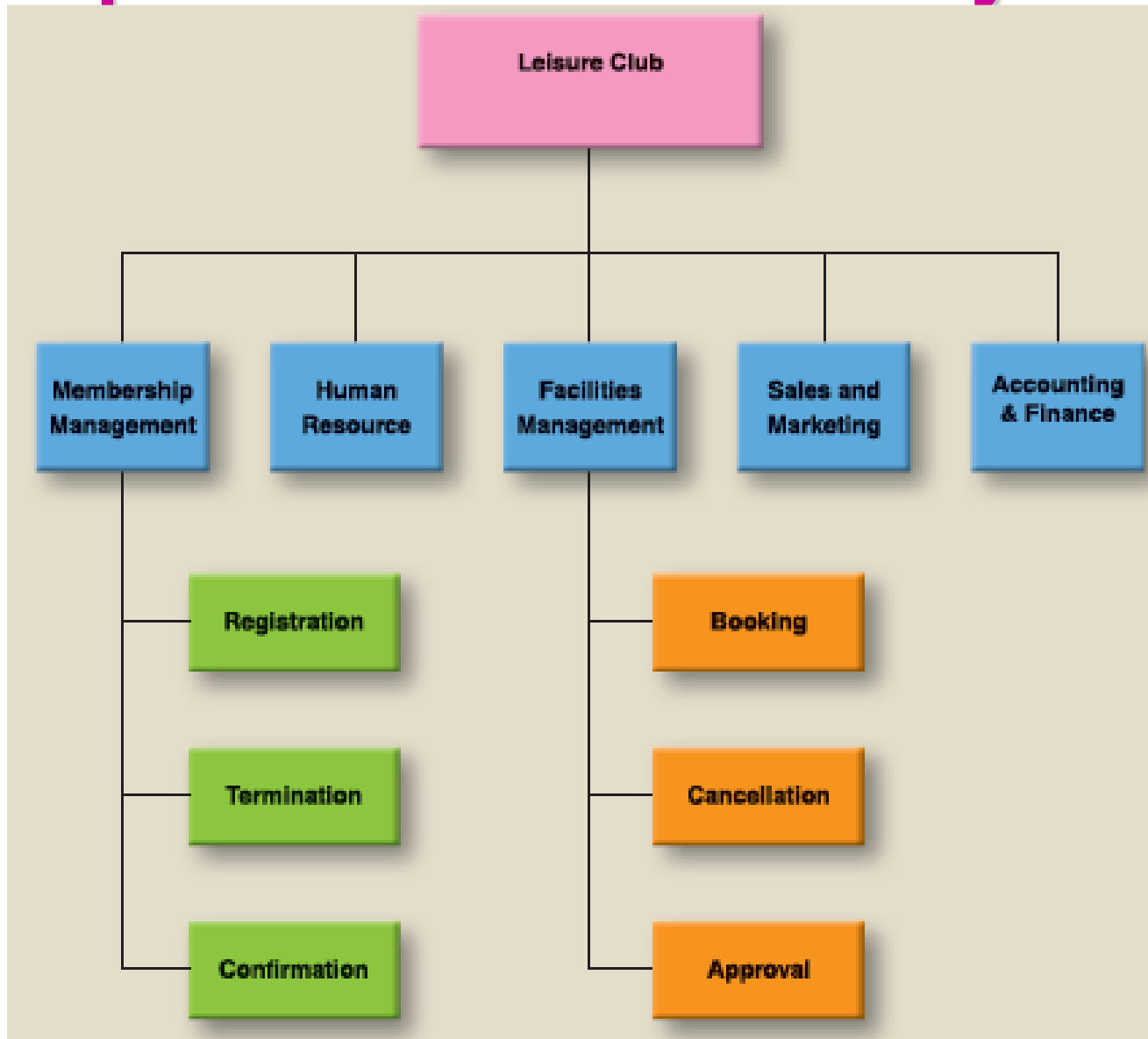


Example 3 – Library System



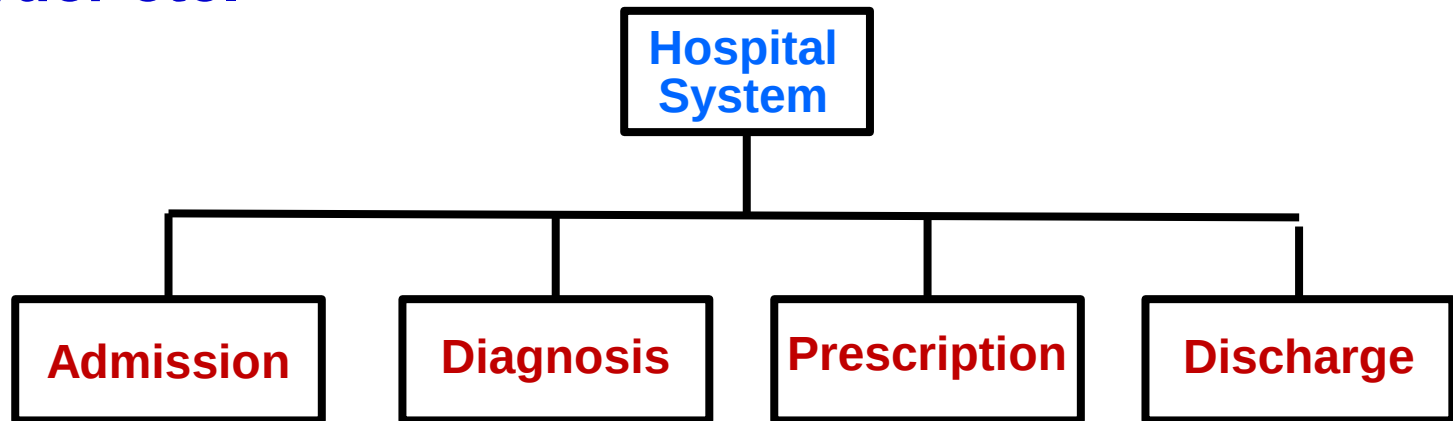
Note : The above is just an EXAMPLE. It may not represent the actual design of any library. **DO NOT** copy this into your Assignment

Example 4 – Leisure Club System



Steps of Decomposition

1. **Identify** the processes eg. Inventory system consists of : Receive Goods, Distribute Goods, Issues Goods, Return Goods.
2. **Group** related activities that need to be performed for each process to form a sub-system eg. for sales system: answer enquiries, create quotation, receive order etc.



User Interface Design

What is a User Interface

- **Meaning**
 - The way users interact with the system when using it.
- **2 Ways :**
 - Hardware
 - Software

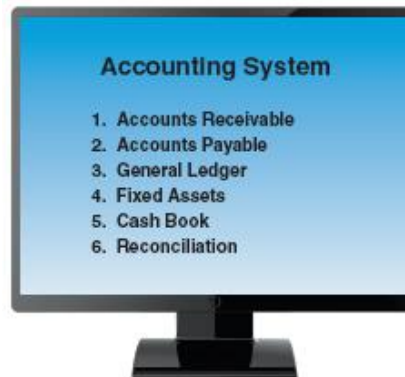


Types of User Interface

1. Command-line Interface



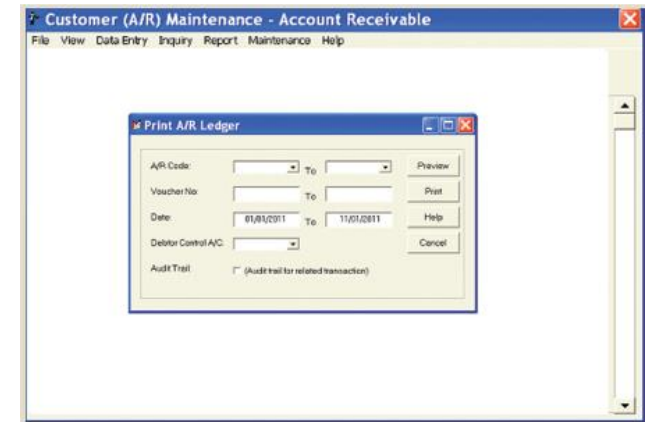
2. Menu-Driven Interface



Types of User Interface

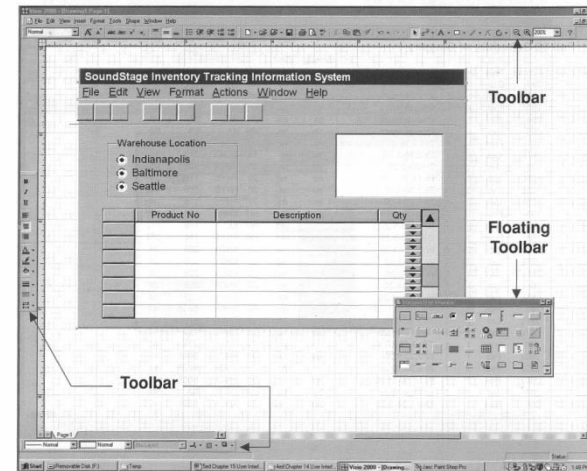
3. Improved Menu Interfaces

- Pop-up Menu
- Pull-down and Cascading Menu



Types of Interface

4. Toolbar and Icon-Based Interface



5. Form-based Interface

Membership Registration

Please fill up all the boxes in the form below and click "Submit" after completed.

Contact Information

Title:

First and Last Name:

Address:

Town or city:

State and Post code:

Country: Malaysia

Phone number:

Hand phone:

Fax number:

Email Address:

Submit

Types of Interface

6. Natural Language Interaction

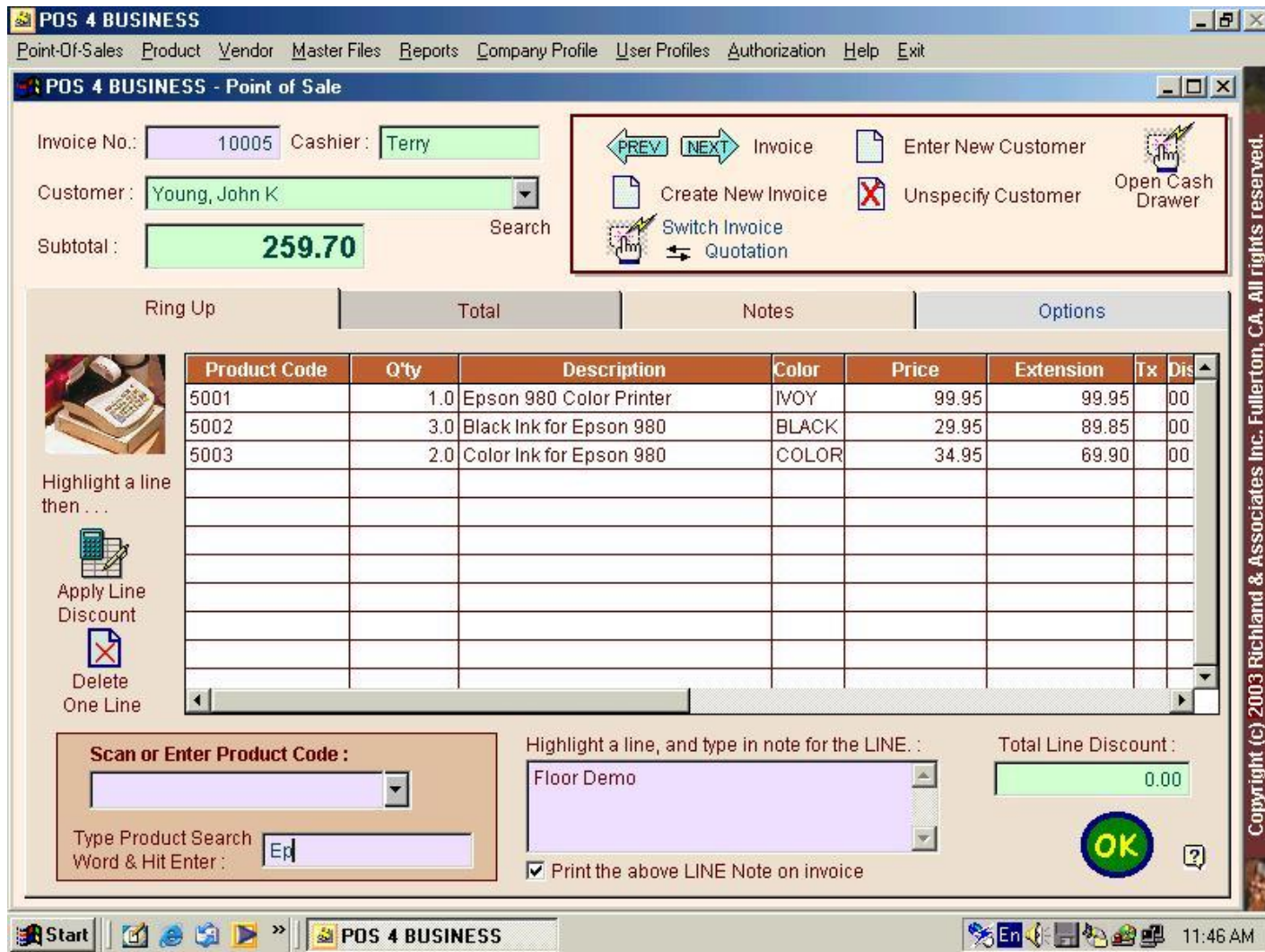
- **English** may be used to interact with the computer system.
- It allows users to **enter** questions and commands, and **produce** outputs in their native language.
- **Examples** - database queries, voice entry systems.

Examples of Input Devices Used

- **Keyboard**
- **Mouse**
- **Touch Screen**
- **Graphics Tablet**
- **Light Pen**
- **Voice**
- **Scanners / Readers**
- **Sensors**

User-Friendliness Characteristics

1. Ease of Data Entry

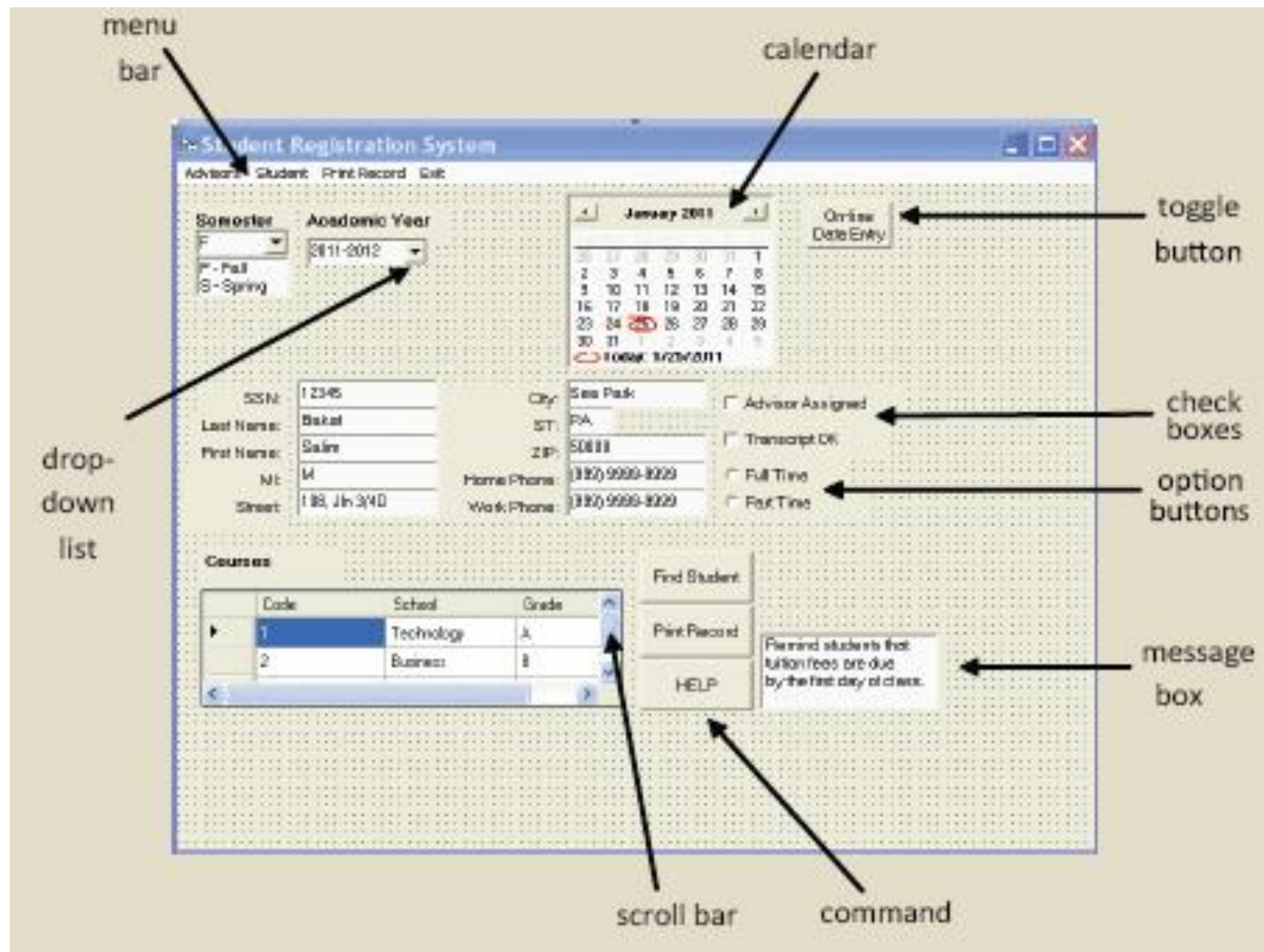


1. Ease of Data Entry

- **Meaning**

- Data entry screen in the same **logical** order as the input form.
- Clearly designed, input fields are highlighted, by **colour, flashing or reverse colouring**.
- Clear titles of fields
- Provision of **data validation checks** in order to reduce potential human data entry errors.

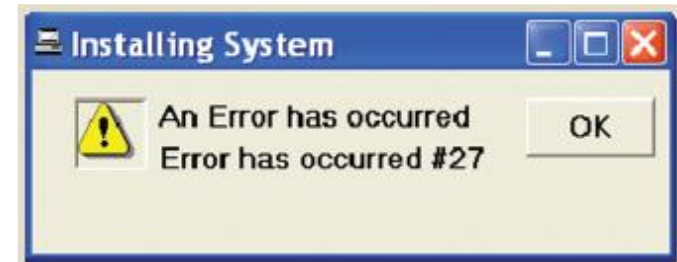
User Interface Controls



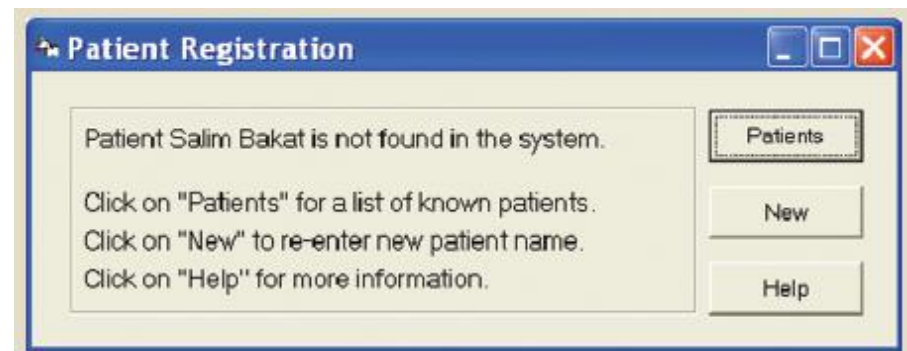
2. Meaningful Error Messages

- **Meaning**
 - All users are likely to make errors.
 - Errors are reported, simple, unambiguous, suggest a course of action.

- **Example of poor error message**



- **Example of good error message**



3. HELP Facilities

- On-screen help
- **Types of Help** : context specific; hypertext and hypermedia.
- **Context Sensitive HELP**
 - **Help** users are unsure to perform a certain function, or made an error.
 - Illustrate with an **example** on the application of the command
 - **Include** a mechanism to allow users to recover from their errors.

4. Consistent with Other Modules

- **Meaning**

- Modules of a system '**look and feel**' and operate in similar way.
- Examples : F1 = 'Help', F2 = 'Save' etc

- **Advantages**

- Facilitate '**transferable**' skills which increase productivity
- Cost saving due to reduced training time

5. Adherence to Industry Standards

- **Meaning** - Use commands and techniques that are used in other software applications.
- **Example** - GUI by Microsoft
- **Advantages** - Facilitate 'transferable' skills which increase productivity users.

6. Escapability (Forgiving)

- **Meaning** - Users may make mistakes when using or learning to use system.
- **Example** – hit wrong keys or type wrong commands eg. delete
- **System response** – Esc or undo keys, message ‘continue or cancel’.

7. Other User-Friendliness Features

- **Pull-Down List**
- **Default Value eg system date, tax rate, discounts, suppliers**
- **Radio Buttons/Checked Boxes**
- **Validation Checks eg. data type check**
- **Automated Inputs eg scanning, reading, sensing**

Pull-down List

- **Meaning**
 - A list of the acceptable or **approved** values that can be selected for a certain data field
- **Examples**
 - Product codes (in order processing system), Male or Female (in HR system).



The screenshot shows a travel booking interface with three tabs: Flight, Package, and Hotel. The Package tab is selected. Below the tabs is a red button labeled "Buy Packages & Save More" with a dollar sign icon. Below this are radio buttons for "Return" (selected) and "One Way". There are input fields for "Origin" and "Destination", each with a pull-down arrow. Below these are "Departure date" and "Return date" fields, each with a date picker showing "21" and "Mar 2007". Below the dates is a "Guests" section with a pull-down arrow. The pull-down list is open, showing a list of months from "Mar 2007" to "Oct 2007". At the bottom of the interface is a red button labeled "Check".

Pull-down List

Others

- **Default Values**
 - Fields entered by the system during data entry. Used when the value of the field can be predicted with some certainty eg. VAT rate, Dates
- **Radio Buttons/Checked Boxes**
- **Validation Checks eg. data type check**
- **Automated Inputs**
 - Scanned, read, sensed inputs

Input Design

Types of Input Devices

- Biometric device
- Data collection device
- Digital camera
- Graphic input device
- Stylus pen
- Keyboard
- MICR
- Microphone
- Mouse
- RFID
- Scanner / OCR
- Terminal
- Touch screen
- Voice input device

Exercise for Students

- **Fine Grocer is a hypermarket located in KL.**
- **Identify 4 areas where input devices are used and what they are in Fine Grocer.**

Suggested Answers

- 1. Receiving in Warehouse –** bar code reader, RFID
- 2. Check-out point –** bar code reader, RFID
- 3. Purchasing dept –** keyboard
- 4. Retail store –** digital camera

Verification and Validation

- **Verification** - Before input data into a system, verification check (eg use human eyes) will be carried out.
- **Validation** - During input, data is subjected to another checks (eg programs) for accuracy before processing could be carried out.
- **Note** – any automated input (ie. not by keying) does not need validation checks.

Types of Data Validation

- **Format Check**
 - Check that data conform to certain expected picture or format.
 - **Example** - a product code have a picture format of AAA/99/AAA.
- **Existence/Presence Check**
 - Check whether the data keyed in exists or not in the relevant database.
 - **Example** - a Social Security number is checked for existence before user is allowed to save the record.
- **Null Value Check**
 - Used to ensure that a value has been entered and not left blank for mandatory fields.
 - **Example** – when filling form, check for mandatory fields
- **Data Type Check**
 - Check that a data item fits the required data type.
 - **Example** - numeric or alphabetic data.

Types of Data Validation (cont..)

- **Size Check**
 - Check that the data conforms to a certain number of characters.
 - **Example** – telephone numbers must be 8 digits.
- **Limit Check** (lower or upper)
 - Check that the data is not below or above a certain value.
 - **Example** – a customer's age must not be less than 10 years (Lower Limit Check).
- **Range Check**
 - Check that the data lies within predetermined acceptable limits (lower and upper)
 - **Example** - in a wages application, check clock card with 'hours worked' outside the range 10-80 hours.

Exercise for Students

- Identify 5 data fields below and suggest suitable validation checks.

Membership Registration

Please fill up all the boxes in the form below and click "Submit" after completed.

Contact Information

Title:		
First and Last Name:		
Address:		
Town or city:		
State and Post code:		
Country:	Malaysia	
Phone number:		
Hand phone:		
Fax number:		
Email Address:		

Submit

Suggested Answers

- Identify 5 data fields below and suggest suitable validation checks.

Membership Registration

Please fill up all the boxes in the form below and click "Submit" after completed.

Contact Information

Title:

First and Last Name:

Address:

Town or city:

State and Post code:

Country: Malaysia

Phone number:

Hand phone:

Fax number:

Email Address:

Data type check

Format check

Size check

Consideration for Input Design

Factors	Comment / Example
Volume	Large volumes require automated input.
Frequency	Infrequent input needs keyboard terminals.
Accuracy	Built-in data validation checks. Automated input.
Speed	Some transactions require that they be input immediately eg. ATM

Example of Input Screen

The image shows a screenshot of a software application window titled "Check-in". The window has a standard Windows-style title bar with minimize, maximize, and close buttons. Below the title bar is a menu bar with "File", "Edit", and "Help" options. The main content area is divided into three sections: "Reservation", "Customer Personal Information", and "Room Information".

Reservation Section: Contains a text input field for "Reservation No." and a "Find Reservation" button.

Customer Personal Information Section: Contains several input fields and controls:

- "First Name" and "Last Name" text input fields.
- "IC No." text input field.
- "Gender" section with radio buttons for "Male" (selected) and "Female".
- "Contact No." section with two text input fields separated by a hyphen.
- "Address" text input field.
- "Post Code" text input field.
- "City" and "State" dropdown menus.
- "Country" dropdown menu.
- "Email" text input field.

Room Information Section: Contains:

- "Check-in date" text input field showing "Tuesday, July 21, 2009".
- "Duration" section with a numeric input field set to "0", a spinner control, and the text "days".
- "No. of Room" section with a numeric input field set to "0" and a spinner control.
- "Room Type" and "Room Number" dropdown menus.

At the bottom of the window are two buttons: "Submit" and "Cancel".

Output Design

Types of Outputs

- **Display** (Screen)
- **Sound** (Sound and audio devices)
- **Printed** (Printers and plotters)
- **Filmed** - Computer output microfilm (COM)
- **Other devices** – Mobile devices, ATM, POS terminals, VDUs

Exercise for Students

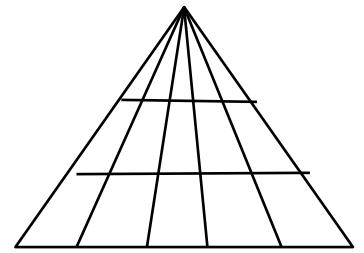
- Identify and briefly explain 3 types of *output devices* used in an airline industry

Suggested Answers

- Printer – for printing of boarding pass
- Speaker – for announcement of departure and arrival
- Screen – for check-in
- Kiosk – for self check-in

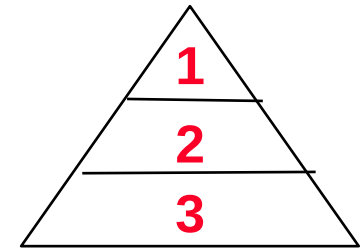
Report Design

Reports Characteristics



- **Functional Area** – a report must be designed for a particular functional area (eg Sales department) eg. Sales Analysis for they Year 2022.
- **Level of Staff/Details** – a report designed for a particular staff of certain level eg. a Summarized Sales for sales manager.
- **Purpose of Report** – a report contains information which are **relevant** and **useful** to the users for making decision

3 Types of Reports



1. Exception reports

- **Meaning** - Filtered items using specific criteria
- **Example**: “Top 5 saleable items for the year 2023.

2. Summary reports

- **Meaning** - Totals and sub-totals, may be graphs or charts
- **Example**: “Total sales for the year 2023, for Item A, breakdown by month, branch, or salesman.

3. Detail reports

- **Meaning**: Report every single item.
- **Example**: Detail daily sales for a customer for the month of Dec, 2023.

Detail Report (sorted and uncategorised)



PIN CHAMPTION INDUSTRIES

No.8, Lorong Perusahaan Ringan Juru 2, 14100 Juru, Pulau Pinang.
Tel: 04-507 1800 Fax: 04-507 1800

19/12/2011

Summary Stock Report on Month December

Material ID	Material Description	Category	Unit of Measurement		Unit Price	Minimum Quantity	Quantity on Hand	Quantity Used
MA00003	Aluminium - 0.7mm	Base Material	48x95	inch	40.00	10.00	349.00	1.00
MA00010	Stainless Steel- 24x48	Base Material	24x48	inch	35.00	10.00	49.00	1.00
MA00011	Stainless Steel - 12x45	Base Material	12x45	inch	25.00	10.00	340.00	1.00
MA00013	Brass - 32x48	Base Material	32x48	inch	23.00	150.00	599.00	1.00
MA00017	Blue Colour	Colour	1000	ml	25.00	1,000.00	77,175.27	1.00
MA00019	Red Colour	Colour	1000	ml	25.00	1,000.00	44,694.99	415.13
MA00023	Green Colour	Colour	1000	ml	25.00	1,000.00	65,423.39	43.01
MA00025	Dark Purple Colour	Colour	1000	ml	25.00	1,000.00	88,537.52	43.01
MA00031	Long Pin	Accessories	2.5	cm	0.10	100.00	4,415.00	362.48
MA00032	Key Chain	Accessories	3	cm	0.15	100.00	2,895.00	20.00
MA00033	Tie Clip	Accessories	3.5	cm	0.20	100.00	3,794.00	5.00
MA00034	Epoxy	Epoxy	3	kg	50.00	3,000.00	268,301.42	853.98
MA00035	Enching	Enching	3	kg	45.00	3,000.00	48,578.78	1,083.62
MA00036	Plastic Bag	Packing	70x50	mm	0.05	100.00	2,422.00	45.00
Total						21,780.00	1,132,839.89	3,059.23

Summary Report

EMPLOYEE HOURS SUMMARY
WEEK ENDING DATE: 6/24/05

PAGE 1

STORE NUMBER	REGULAR HOURS	OVERTIME HOURS	TOTAL HOURS
8	181.2	13.5	194.7
11	184.8	23.5	208.3
17	172.0	19.4	191.4
	—	—	—
TOTALS:	538.0	56.4	594.4

Exception Report

OVERTIME REPORT
WEEK ENDING DATE: 6/24/05

PAGE 1

STORE NUMBER	POSITION	EMPLOYEE NAME	OVERTIME HOURS
8	Asst Mgr Manager	Davenport, Kim	5.0
		Ramirez, Rudy	8.5
	STORE 8 TOTALS:		13.5
11	Manager	Gadzinski, Barbara	10.0
	Clerk	Stites, Carol	12.0
	Asst Mgr	Thompson, Mary Kay	1.5
STORE 11 TOTALS:			23.5
17	Clerk	De Martini, Jennifer	8.4
	Clerk	Rittenbery, Sandra	11.0
STORE 17 TOTALS:			19.4
GRAND TOTAL:			56.4

Reporting OT hours for **certain** Stores

Report Design Layout

Column Space

PiN **PIN CHAMPTION INDUSTRIES**
No.8, Leong Permashan Ringan Juru 2, 14100 Juru, Pulau Pinang.
Tel: 04-507 1800 Fax: 04-507 1800

Sales Report Month December

Person incharge: Ang Ming Yong

Date	Memo Number	Memo Description	Amount
2011-12-29	PM00003	custome 2	41.26
Total			41.26

Person incharge: Lee Hoay Ling

Date	Memo Number	Memo Description	Amount
2011-12-29	PM00004	Sample D	33.44
2011-12-29	PM00005	Sample E	33.82
2011-12-29	PM00007	testing	157.20
Total			224.46

Person incharge: Ng Hooi Ming

Date	Memo Number	Memo Description	Amount
2011-12-29	PM00001	Sample A	48.00
2011-12-29	PM00002	Custome A	90.15
Total			138.15

Person incharge: San Lin Xin

Date	Memo Number	Memo Description	Amount
2011-12-29	PM00005	Sample C	31.45
Total			31.45

Total Amount			435.32
--------------	--	--	--------

Report Header

Page Header

Record Group

Group Totals

Report Footer

Attributes of a Good Report

- **Appropriate Report Title**
 - **Name.** The report title should lead the readers to know what it will contain.
 - **Length.** The title used should be clear, simple and concise.
- **Pleasing Report Layout**
 - **Labels.** All data entry fields and data should have appropriate labels.
 - **Layout.** Information on a report should be well layout on paper or displayed on a screen with sufficient margins and spaces between the data items.
- **Meaningful Report Content**
 - **Presentation.** Information contained in a report should be presented in a way that is useful to the users.
 - **Relevant.** A report should only contain information which are relevant and required by the users.